
The Price of a Flat Direction: Transverse Curvature Sets Retention in a Trained Recurrent Memory

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Abstract

1 A recurrent state can store a continuous value on a manifold of marginally stable
2 states. This literature already knows that such a manifold is fragile: noise diffuses
3 the state along it, most infinitesimal perturbations of the dynamical law destroy it,
4 and corrective mechanisms that keep it alive are classic. It also knows that exact
5 equivariance protects a neutral direction: a recent preprint (arXiv:2605.03338)
6 proves that an exactly equivariant field carries zero Lyapunov exponents tangent
7 to the group orbit, and reports that in its controlled breaking experiments the
8 formerly protected direction predicts a finite memory lifetime. *What has not been*
9 *available is the exchange rate.* We measure one, on a trained potential. The
10 object is a recurrent unit advancing a latent state (q, p) by a damped symplectic
11 velocity-Verlet step of a learned Hamiltonian. Its trained potential V_θ is G -invariant
12 with minimisers on a G -orbit — spontaneous symmetry breaking — so that the
13 curvature μ transverse to that orbit prices how long a written value survives. The
14 price list has three parts, every number at latent dimension 4 on an S^1 testbed, ≤ 5
15 seeds, laptop CPU. Retention half-life scales as $n_{1/2} \propto \mu^{-2}$, at fitted overdamped
16 slope -0.985 against a predicted -1 , the underlying curvature identity exact to
17 $1.000000 \pm 5 \times 10^{-12}$ over 4.5 decades. The law stops at a crossover $\varepsilon\mu \approx \gamma/2$,
18 past which the half-life saturates at a curvature-independent floor. And the whole
19 list survives the training-time correction that keeps the orbit alive, at $\approx 20\times$
20 the horizon on which the objective would otherwise destroy it. Its evidence on
21 a learned system is a head-to-head: that preprint’s single-exponential lifetime
22 estimator, run unchanged on our trained checkpoints, is this price list’s overdamped
23 face — matching to $\leq 3\%$ below the crossover and mispredicting by $\approx 3.2\times$ above
24 it, in the calculable direction. *We therefore also report where the price list does*
25 *not extend, and those boundaries are part of the result:* on this architecture class
26 symmetric training data alone does not produce a near-flat direction, a gap of 13–14
27 orders in μ^2 that is a measurement on our recipe and never a general claim about
28 learning; the continuous coset register is designed-only; the raw retention ratio
29 against recurrent baselines inverts under compute normalization and is retired as a
30 compute claim; and four positioning claims are retired against prior art. Designed-
31 testbed results are labeled verification; learned-system results, evidence.

32 1 Introduction

33 A recurrent state holding an analog value can be built on a flat direction: a set of states the dynamics
34 do not distinguish, so a written value neither decays nor is pulled to a preferred point. Symmetry
35 supplies such directions for free, and it supplies them by breaking. If a learned potential V_θ is
36 G -invariant and its minimiser is not — spontaneous symmetry breaking, stated in the ordinary sense

— the minimisers form a G -orbit, the Hessian annihilates every orbit tangent, and the orbit coordinate is neutral. That neutral coordinate is the Nambu–Goldstone mode of the trained potential, and it is the register this paper prices. *However*, a flat direction is only as useful as its price list, and none existed for a trained model: how much retention a given transverse curvature buys, where that law stops holding, and whether it survives the correction that keeps the orbit alive. *We measure all three.* The object advances (q, p) by a symplectic velocity–Verlet step of a learnable separable Hamiltonian $H(q, p) = T(p) + V_\theta(q)$, with a per-step damping $p \mapsto (1 - \gamma)p$ supplying controllable forgetting. This is not a new architecture but a closed-form account of retention on a trained one. The reference unit is the CLU (Causal Learning Unit), introduced as CHLU in Jawahar & Pierini (2026); results hold for the class of damped symplectic recurrences, and the exactly-solvable theory verified here is developed in a separate note, cited as *the theory note* (Anonymous, 2026, in the supplementary material).

Two masses, and one conversion. Two quantities here are called “mass” and run in opposite directions. The inertial mass M_i is the learned diagonal of T : larger M , slower. The spectral mass μ_k , which we lead with as the *transverse curvature*, is $\mu_k^2 = \lambda_k(M^{-1/2}\nabla^2 V_\theta M^{-1/2})$ at a critical point: larger μ , shorter memory. All retention statements use μ . One conversion precedes any cross-field comparison, and it is not a rescaling: flow-Jacobian eigenvalues — Lyapunov exponents — are inverse time, whereas μ^2 are eigenvalues of a mass-whitened Hessian of a *potential*, inverse time squared. We therefore never convert a rate into a curvature on paper; §4.2 does the conversion the only way we trust, by running a published rate-based estimator unchanged on our trained models. What this literature calls a *memory lifetime* is our half-life $n_{1/2}$, reported in map applications, not τ and not diffusion coefficients D ; each is reported with the metric defining it, because those metrics bifurcate past the crossover.

Contributions. This paper makes one contribution, and it is quantitative: *the transverse-curvature price list of a trained recurrent memory*. It is delivered in three pieces.

1. *The price list* (§4.1, §4.4), in three parts: what a given transverse curvature buys, where that law stops, and whether it survives the training-time correction that keeps the orbit alive.
2. *Its evidence on a learned system* (§4.2): a recently posted single-exponential lifetime estimator, run unchanged on our trained checkpoints, is the price list’s overdamped face and fails above the crossover in the direction the list predicts.
3. *Its honest negative* (§4.3): on this architecture class the flat direction has to be designed rather than trained for, the continuous register does not transfer to an emergent potential, and the retention advantage over recurrent baselines inverts under compute normalization.

Designed-testbed results — invariant potentials, analytic tilts and spurions — are verification of the theory’s exactness; learned-system, training-dynamics and head-to-head results are evidence.

2 Related work

Two questions this audience is asking — how neural representations behave *dynamically*, and what symmetry buys a learning system — meet in one object, and this paper prices that object. Neutrality of group-orbit directions is classical (Golubitsky, Stewart & Schaeffer 1988; Krupa 1990; Rumberger 2001). The same object is studied as a *continuous attractor*: a manifold whose points are all fixed points, “marginally stable tangent to the manifold and stable normal to the manifold” (Ságodi et al. 2024), the substrate of the neural integrator (Seung 1996) and of measured ring and toroidal codes (Kim et al. 2017; Gardner et al. 2022; Khona & Fiete 2022). We make no claim about biological systems and do not model neural data.

Five facts about that object are established, and none of them is ours. Noise diffuses the coordinate along the manifold, the diffusion coefficient lower-bounded by the code’s own Fisher information (Burak & Fiete 2012). The manifold is structurally unstable, destroyed by most infinitesimal changes of the dynamical law, with online learning named as one perturbation source — the *fine-tuning problem*, whose memory error Ságodi et al. bound by the unknown perturbation’s size, worst-case and linear in t . A homeostatic mechanism that restores a destroyed flat direction is 23 years old (Renart, Song & Wang 2003), so a corrective term is not our idea, and nothing here is a first report

of the destroy-and-restore pattern. A local learning rule can *produce* accurate ring-attractor tuning (Vafidis et al. 2022), so nothing we measure implies that learning cannot build a flat direction. And approximate continuous attractors arise in other task-trained recurrent networks by themselves, with a recent analytical model of the slow structure appearing during training (Dinc et al. 2025, preprint), where §4.4 runs the opposite direction. *Absent is the exchange rate: how much retention a given transverse curvature buys, where that law stops holding, and whether it survives the correction. We measure all three on trained models, in closed form, with a regime map — floor, crossover — that a first-order dynamical system cannot exhibit.*

The nearest current work, and what it does to us. The closest live line to this paper carries a one-parameter Lie-group symmetry through time in a recurrent state, under the name *flow equivariance* (Keller 2025), and extends it to world models that hold a latent coherent over long horizons of partially observed motion (Lillemark et al. 2026). *One disambiguation, because this literature uses the word three ways: flow here means a one-parameter Lie subgroup acting through time, never Ricci flow and never a normalizing flow.* That line and this paper share an object — a continuous symmetry carried in a recurrent state over a long horizon — and split the question cleanly: *they generalize, we price.* Flow-equivariant models are shown to extrapolate to unseen velocities and beyond the training horizon; we measure what a given transverse curvature costs in retention, and make no generalization claim of any kind. The soft counterpart of our designed arm is also current: *Symmetry-regularized learning of continuous attractor dynamics* (NeurReps 2025 workshop poster) obtains the flat direction by symmetry regularization of a learned model, where our designed arm obtains it by construction and our emergent arm carries no symmetry regularizer at all, which is exactly the scope of the negative in §4.3. On the training-dynamics side the organizing question is how learning shapes attractor geometry: geometric restructuring precedes abrupt learning in short-term-memory tasks, and a temporal-consistency regularizer facilitates attractor formation (Haputhanthri et al. 2025, preprint). §4.4 runs that arrow backwards — our objective *destroys* a designed flat direction and an anchor holds it up. Finally, the standard instrument for the kind of gap we report between a designed and an emergent recipe is solution degeneracy across task-trained recurrent networks (Huang, Singh, Martinelli & Rajan 2025); we do not run it here, and §5 names it as a direction.

What is already established about protected directions, and what is not. A recent preprint on symmetry-protected Lyapunov neutral modes (arXiv:2605.03338), the equivariant-Lyapunov preprint below, proves that a C^1 field exactly equivariant under a Lie group has at least $\dim(G/\mathcal{H})$ zero Lyapunov exponents tangent to the orbit, and reports that in its controlled breaking experiments the formerly protected direction predicts a finite memory lifetime, on S^1 path integration against matched gated and orthogonal recurrent baselines. *That existence result, and that qualitative lifetime prediction, are established there and are not claimed here.* What this paper adds is the exchange rate on a trained potential, measured at latent dimension 4 on an S^1 testbed on ≤ 5 seeds: the closed-form price list of §4.1 — the μ^{-2} law with its measured slope, the curvature-independent floor, the exceptional-point crossover, and the survival of all three under the corrective anchor. The currencies differ, and the difference is load-bearing: its gap is a Lyapunov exponent, an inverse time, while our μ^2 is a curvature of a trained potential, an inverse time squared. That result is the *kinematics* of protection, silent on what sets the gap; here the gap is *constitutive*, and Lyapunov spectra are blind to the write semantics — transport, latch, noise accumulation — that make a neutral direction a *memory*. The theorem states sufficiency; equivariance is not necessary, the latch being a modulus of V_θ . We use its diagnostics and its breaking-and-censoring protocol, and §4.2 runs its estimator unchanged on our trained models.

Closest to our object among this audience’s own earlier work is Xu et al. (2023), where a Lie-group representation is carried in the recurrent state of a grid-cell network and the compact, connected, abelian representation forces the torus topology over a continuous attractor. That is our geometry, in a network that path-integrates; what we add to it is a price list for time. Two further neighbours ask the same question in different currency: the optimal packing of attractor states in a representation, where symmetries of the environment’s transition statistics imply symmetries of the optimal code (Vastola 2024), and memory modification approached through symmetry and geometry (Dönmez 2024). Our designed-versus-emergent axis is this audience’s exact-equivariance-versus-symmetry-breaking axis (van der Ouderaa & van der Wilk 2023; Iqbal, Keller, Song, Miyato & Welling 2026). Reading a landscape by its level sets and critical points is likewise native here (Akhtiamov & Thomson 2023; Wang & Ponce 2023, whose level sets are of tuning landscapes over *stimuli*, not of a latent-state

potential — the two must not be conflated). A canonicalization, a coordinate on the quotient, has been used as an input encoding (Aslan, Platt & Sheard 2023), where ours is a storage device. Constructive equivariant memory (Di Bernardo et al. 2025) engineers *which* symmetry a network expresses but does not price *time*: we claim no novelty for choosing a symmetry group or coset, only the price list attached to a chosen structure. Cross-session representational drift (Driscoll et al. 2017; Aitken et al. 2022; Jude et al. 2023) is a different phenomenon from motion along a flat direction and we do not address it; the only motion we report is of the coset coordinate under our own dynamics. Our comparators are LSTM (Hochreiter & Schmidhuber 1997), LEM (Rusch et al. 2022) and coRNN (Rusch & Mishra 2021a); conformal symplecticity is McLachlan & Perlmutter (2001), the leapfrog $h < 2$ limit standard (Hairer, Lubich & Wanner 2006), and wake-sleep (Hinton et al. 1995) with PCD (Tieleman 2008) the contrastive objective, whose landscape distortion is classical (Fischer & Igel 2010; Nijkamp et al. 2020).

Four claims we retire, so that a reviewer’s first objections are answered in writing (elaborated, with their numbers, in Appendix F). (1) Attention and state-space models *do* have retention-guarantee analogues, and HiPPO-LegS (Gu et al. 2020) is the counterexample. (2) “Retrieval is a rollout, not a weighted sum” is Kong, Brewer & Lai’s (2024) move, not our novelty. (3) Continuity is no escape hatch from the NTM/DNC lineage (Graves et al. 2014, 2016; Csordás & Schmidhuber 2019), which was already fully soft. (4) This is no competitor to the transformer, and was never claimed to be. Our claim is correspondingly narrow: a physics-structured associative memory in which retention is a designed, per-item property — permanent (symmetry-protected, $\mu^2 \equiv 0$ exactly) or of a finite half-life set by μ^2 — rather than an emergent consequence of a learned recurrence — *controllability*, which we evidence, not *capability*, which we do not claim against transformers or state-space models. The nearest published neighbour to our write is Titans (Behrouz et al. 2025), test-time gradient descent with momentum on an associative-memory loss: the same family of discretized damped second-order dynamics, differing in that our damping and inertia are *read out and priced*, not optimizer hyperparameters.

3 Setup

The map, and the three bands. One dissipative velocity-Verlet step with step ε and damping γ advances (q, p) by a kick-drift-kick of $H = T(p) + V_\theta(q)$ followed by $p \mapsto (1 - \gamma)p$. It is conformally symplectic — $J^\top \Omega J = (1 - \gamma)\Omega$, $\det J = (1 - \gamma)^d$; phase-space volume contracts by a fixed factor per step, so the geometry is preserved and only the scale is spent. Memory at each normal mode reduces to one 2×2 block ($\mu^2 = k/m$, $h = \varepsilon\mu$) with three bands: a latch ($\mu = 0$, symmetry-protected) with frozen displacement $q_\infty = q_0 + \varepsilon p_0/(M\gamma)$ and infinite half-life; an overdamped register ($0 < \varepsilon\mu \lesssim \gamma/2$) at $n_{1/2} \approx 2\gamma \ln 2/[(2 - \gamma)(\varepsilon\mu)^2]$; and an underdamped working memory ($\gamma/2 \lesssim \varepsilon\mu < 2$) at the curvature-independent envelope half-life $2 \ln 2/(-\ln(1 - \gamma))$. The boundary is the exceptional point $h^* \approx \gamma/2$, the overdamped-to-underdamped crossover at which the block becomes defective. Relaxation onto the orbit is the fast normal flow; retention is the slow tangent flow. One difference from a classical neural integrator we do not smooth over: an integrator stores by *not* damping, whereas at $\gamma = 0$ our flat mode never freezes a write and any $\gamma > 0$ gives an exact latch.

What the curvature can be, and what the unit stores. At a critical point of V_θ the transverse curvature takes one of three values, and trained models here realize all three: *exactly* zero along $\dim(G/\mathcal{H})$ directions when the trained minimisers form a G -orbit, the spontaneously broken case (a Nambu-Goldstone mode — the designed unit, and the exactly-flat-direction geometry); small but nonzero when an explicit tilt breaks the orbit, giving a slow manifold rather than a flat one (a pseudo-Goldstone mode — the *explicitly-broken cell*, which the analytic tilts of Appendix G instantiate and the self-broken MLP exhibits); and strictly positive with no orbit of minimisers at all when the minimiser is itself invariant, degenerate within each irrep by Schur’s lemma (the contrastive-divergence-eroded vacuum of §4.4). Only the *classical, tree-level* Goldstone theorem is used, a statement about a Hessian. No thermodynamic limit, no phase transition, no $\hbar$, and no loops, chiral logarithms, running low-energy constants or anomalies anywhere in this paper. What the unit stores is the *coset coordinate*, the position along the flat direction — a canonicalization used as a storage device rather than as an input encoding; the map’s (ε, γ, T) fixes what a given μ^2 buys, through the three bands above.

198 **Trained models.** All checkpoints are the reference unit’s Experiment-D configuration — dim 4,
 199 hidden 64, learned-Newtonian kinetic term, $dt = 0.05$, unit circle at 256 points, 150 epochs (§4.4 for
 200 why not the default 1000) — in two families: designed, an architecturally- $SO(2)$ -invariant V_θ on 5
 201 seeds, and emergent, a generic MLP V_θ on the same data on 3 seeds.

202 *Fine print, next to the claims it qualifies.* (a) The latch’s infinite half-life is a $T = 0$ statement about
 203 the deterministic damped map; at $T > 0$ the coset coordinate diffuses with $D_\theta = \varepsilon T(2 - \gamma)/(2F^2\gamma)$,
 204 $F^2 = M_{\text{ch}} r^{*2}$ — a *diffusive register* of finite computable lifetime, verified on the trained designed
 205 checkpoints, where raising friction *lengthens* coset memory by $3.77 \pm 0.23\times$ for $\gamma : 0.05 \rightarrow 0.2$
 206 on 5/5 seeds, and holding only under fluctuation–dissipation-consistent noise and a Newtonian
 207 kinetic mode, never under the reference implementation’s legacy noise default. All §4 results
 208 are $T = 0$. (b) By Sylvester’s law of inertia a flat direction keeps $\mu^2 = 0$ under any inertial
 209 anisotropy: symmetry protects the write *current*, flatness protects the *register*, and map equivariance
 210 is sufficient but not necessary for a neutral memory direction — our kinetically-broken battery is a
 211 *measured* non-equivariant map that latches exactly (Appendix E). (c) The *tilt-as-a-designed-lifetime-*
 212 *dial* construction is claimed for designed geometries only, the self-broken MLP being an *observed*
 213 instance of the cell and not a dial: in a separate battery of ours on a *learned, multi-atom* store, a
 214 breaking coefficient intended as the lifetime dial moved the softest eigenvalue *the wrong way*, and a
 215 coercivity term — not the breaking — set the ceiling (Appendix E).

216 *Mandatory flag, travels with every finite-temperature number in this paper.* All such results
 217 require `langevin_noise="fdt"` — the fluctuation–dissipation-consistent noise scale $\sigma_i^* =$
 218 $\sqrt{M_i T \gamma (2 - \gamma)}$ — and a Newtonian kinetic mode (`newtonian_identity/newtonian_learned`).
 219 The reference implementation’s default is `legacy`, under which T is not in energy units and none of
 220 these laws hold; and in the relativistic kinetic mode no noise scale targets Gibbs at all (Appendix E; *the-*
 221 *ory note*, Prop-9’), so the laws are undefined there. The checkpoints here are `newtonian_learned`,
 222 so every finite-temperature number here is in scope. Nothing in §4 depends on it: all main-text
 223 retention results are $T = 0$ deterministic rollouts.

224 4 Results: the price list, its evidence, and its boundary

225 4.1 What a curvature buys, and where the law stops

226 A zero eigenvalue certifies that a direction is neutral. *It does not say what a nearly-zero one costs. We*
 227 *therefore tilt a trained potential by a known amount, read the resulting transverse curvature off its*
 228 *Hessian, and measure the half-life of a written value.* On the trained designed vacuum, whose flat
 229 mode has $\mu^2 \leq 2.4 \times 10^{-15}$ and is machine-flat, the constitutive identity holds: the Hessian-derived
 230 spectral mass equals the measured one-step-Jacobian gap to 3.2×10^{-10} over all 70 tilt rows. The
 231 retention law follows, with the curvature ratio $\mu_{\text{meas}}^2/\mu_{\text{pred}}^2 = 1.000000 \pm 5 \times 10^{-12}$ at every tilt
 232 and seed across 4.5 decades and a fitted overdamped slope of -0.985 ($n = 35$; predicted -1). Past
 233 the crossover the half-life stops depending on curvature at all, saturating at the floor 27.03 steps
 234 at $\gamma = 0.05$. Two second-order signatures confirm that the crossover is a genuine exceptional —
 235 defective — point and not a fitting artifact: the retention metric bifurcates there, envelope half-life
 236 and first-crossing time agreeing to 0.01% overdamped and splitting by $3.2\times$ at the deepest tilt, which
 237 is why every lifetime in this paper names its metric; and the oscillation frequency turns on with the
 238 predicted square root, $\varphi = 0$ exactly below h^* (15/15 rows) and $\varphi \propto \sqrt{h - h^*}$ above it at fitted
 239 slope 0.5165 against the predicted 1/2. The latch corner of the same block is the transport law
 240 $q_\infty = q_0 + \varepsilon p_0/(M\gamma)$, which holds to $\leq 1\%$ over two decades of γ , the written angle then frozen to
 241 $\leq 1.2 \times 10^{-15}$ rad. *These are trained checkpoints carrying analytic tilts: verification of the theory’s*
 242 *exactness on a learned potential, not a discovery.* Appendix G records the battery in full, the probe’s
 243 own roundoff floor, and the damping corollary that makes $n_{1/2}(\gamma)$ non-monotone.

244 4.2 Evidence: a published lifetime estimator is the price list’s overdamped face

245 The price list should predict what other instruments see. *A published estimator gives the direct test,*
 246 *because it was fitted to a different object. We therefore run it unchanged on our trained models*
 247 *and read where it holds.* The equivariant-Lyapunov preprint (arXiv:2605.03338) reports that, in
 248 its controlled breaking experiments, a broken protected direction predicts a finite memory lifetime
 249 through a single-exponential estimator of its own; that prediction is its result, not ours. Its estimator

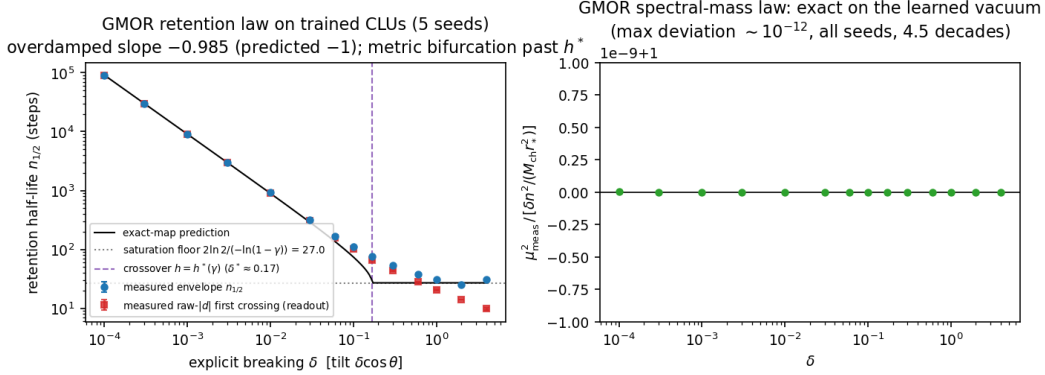


Figure 1: The price list, verified on trained checkpoints. Left: the retention half-life $n_{1/2}$ against the analytic tilt δ that sets the transverse curvature. The overdamped branch follows the predicted law at fitted slope -0.985 against a predicted -1 and then saturates, past the crossover $h = h^*(\gamma)$, at the curvature-independent floor 27.03 steps at $\gamma = 0.05$; the two retention metrics — envelope half-life (circles) and raw first-crossing time (squares) — agree through the overdamped band and separate underdamped, by up to $3.2\times$ at the deepest tilt, which is the exceptional point’s second-order signature and the reason every lifetime in this paper names its metric. Right: the underlying curvature identity, measured over predicted spectral mass, flat at $1.000000 \pm 5 \times 10^{-12}$ across 4.5 decades of δ . *Trained designed checkpoints carrying analytic tilts, 5 seeds, dim 4, $\gamma = 0.05$, laptop CPU: verification of the theory’s exactness. The full battery, the probe’s own roundoff floor and the damping corollary are in Appendix G.*

is built on a Lyapunov exponent, an inverse time, and ours on a curvature, an inverse time squared, so the two are compared here by running its instrument rather than by converting its units. Running its exact code-level protocol — phase 0.35 rad, threshold 0.2 rad, its censoring, cap 15000 steps — unchanged on the 5 trained models across 14 breaking magnitudes gives measured/predicted $1.012 \pm 0.000 \rightarrow 1.029 \pm 0.001$ overdamped against its reported median 1.013, 2.202 ± 0.155 at the exceptional point, and 0.309 ± 0.012 deep underdamped. That estimator is the overdamped face of our price list: there the two coincide, $\text{corr}(\log \text{pred}, \log \text{meas}) = 0.9987$ overdamped-only, and its predictor fails only past the crossover — by $\approx 3.2\times$ at the deepest trained-model breaking we test, $\delta = 4$, our sweep going no deeper underdamped — in the calculable ballistic direction, with the $2.2\times$ exceptional-point delay spike as the third signature. On the *exact analytic map* the same mechanism continues to $\approx 5\times$ further underdamped (theory note, check (k), $\gamma = 0.1$); the $\approx 3.2\times$ is the trained-model number, the $\approx 5\times$ is exact-map only, and we never quote it as evidence about a trained network. This is containment, not conflict: below h^* we recover its reported number, and above it the regime structure — floor, ringing, exceptional point — is what a first-order dynamical system cannot exhibit and what a *constitutive* curvature adds. It also does not rest on substituting the exact Jacobian gap for its predictor: the result reproduces on the preprint’s own finite-horizon estimator (Appendix F). *Evidence: a recently posted ML lifetime estimator, run on trained checkpoints.*

4.3 The honest negative: where the price list does not extend

Symmetric training data might be expected to induce a flat direction on its own. *On this architecture class it does not.* A generic MLP potential trained on that data develops *no* near-flat direction: softest emergent $\mu^2 = 5.1/5.9/5.4 \times 10^{-2}$ on 3 seeds against the designed flat mode $\leq 2.4 \times 10^{-15}$, a gap of 13–14 orders; the preprint’s own attribution instrument separates the architectures by 15 orders. This is a measurement on our architecture class and training recipe, not a general statement that learning cannot produce a tuned flat direction — a local learning rule that does produce one is published (Vafidis et al. 2022). Operationally that arm stores nothing on a *continuum*: ≈ 1 –1.6 bits on 2–3 washboard minima, a written δ relaxing completely, though the damping-optimum retention law itself still generalizes to it. The continuous coset register is therefore designed-only, and we mark it as the one face of the price list that does not transfer.

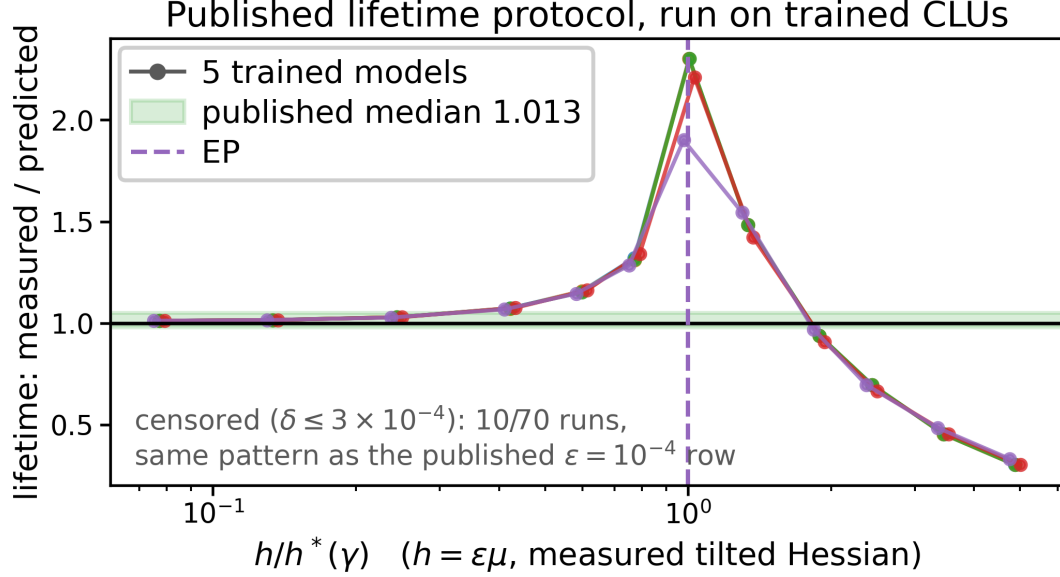


Figure 2: The single-exponential lifetime estimator of the equivariant-Lyapunov preprint (arXiv:2605.03338), run *unchanged* on the trained models, is the overdamped face of the transverse-curvature price list. Measured/predicted lifetime ratio versus breaking magnitude, 14 regimes, 5 seeds: ≈ 1 overdamped, reproducing its reported median 1.013; a $2.2\times$ delay spike at the exceptional point; declining to $0.31\times$ deep underdamped — containment exact below the crossover, failing above it in the calculable ballistic direction. *Evidence.*

278 **The same boundary against learned baselines.** Gated and oscillatory recurrent networks fit this
 279 task well. *Fitting it is not holding it. We therefore write a value into each and let it run autonomously.*
 280 On the preprint’s S^1 path-integration family the baselines are trained with a learning-rate sweep, best
 281 RMSE kept, and are not strawmen: LSTM and LEM reach train-horizon RMSE 0.18/0.23 rad. Yet
 282 under its *autonomous* protocol the structural triad of latch, μ^{-2} law and bounded motion of the stored
 283 coordinate is qualitatively absent in them: coRNN/LEM/LSTM forget the stored analog phase in
 284 $\approx 5.6/56/69$ map-steps, the stored coordinate randomizes past 1.2 rad, and they are *perturbation-*
 285 *fragile*, a 0.1 hidden kick collapsing LSTM $69 \rightarrow 2$ and LEM $56 \rightarrow 5$, while the generically-trained
 286 unit holds ≈ 263 map-steps with bounded motion of that coordinate (≈ 0.35 rad, price-list law to
 287 $+12\text{--}15\%$) and the designed unit never moves (5/5 seeds). *Evidence.*

288 **Honest gap.** The load-bearing claim is the qualitative triad, architecture- and compute-independent.
 289 The raw 263-versus-69 map-step ratio does not survive compute normalization, and we retire it
 290 as a compute claim: per step the Verlet update costs $\approx 6.2\times$ an LSTM cell and $\approx 3.1\times$ a LEM
 291 cell in wall time, $\approx 14\text{--}15\times$ FLOPs, not width-matched, so per unit of retention the unit spends
 292 $\approx 23.5\times/\approx 14.6\times$ more wall time than the baselines it out-holds (Appendix D). Lacking native
 293 velocity ingestion, it cannot enter the preprint’s input-driven task-RMSE axis at all, so no task-RMSE
 294 was fabricated.

295 4.4 The price list survives the training-time correction

296 A price list read off a potential is worth little if the training objective quietly rewrites the potential.
 297 *Ours does exactly that.* Wake–sleep contrastive divergence drives the order parameter r^* to zero,
 298 inverting the designed vacuum in 8/8 runs at the default 1000 epochs: the minimiser becomes
 299 invariant, with the degenerate μ^2 pair predicted in §3, on a horizon set by sleep-update *frequency*
 300 racing the wake clamp, and specific to flat or degenerate vacua. This is an instance of what the attractor
 301 literature calls the fine-tuning problem, the perturbation here being the training objective itself.
 302 *Scope clause, travelling with that finding:* our sweep varies chain length only over `sleep_steps` $\in$
 303 $\{50, 500\}$ at otherwise fixed sampler settings and does not resolve where either sits relative to this
 304 model’s mixing time, so the finding is *frequency-decisive across the two chain lengths we run*, and is
 305 not a claim that chain length is irrelevant to contrastive-divergence learning in general.

306 A $V(\text{data})$ energy anchor holds the condensate up — $\lambda = 100$: 5/5 seeds, $r^* = 0.911 \pm 0.016$, at
307 the cost of weaker noise rejection and $\approx 35\times$ higher wake MSE. *That a corrective term can keep*
308 *a flat direction alive is not new* (Renart, Song & Wang 2003). What we add is the third part of the
309 price list: every headline law of it still holds under the correction, at $\approx 20\times$ the erosion horizon
310 (Appendix A, Figure 3) — 3000 anchored epochs, 3 seeds, curvature ratio exact to 1.5×10^{-12} over
311 4.6 decades, slope -0.956 , the same floor, and the exceptional-point onset bit-identical at 0.5165.
312 The architecture *breaks* the symmetry, the objective *restores* it, the anchor *keeps it broken*.

313 5 Discussion

314 *Scope box — limitations and scope choices, stated so no reader has to infer them.* All results are
315 laptop-CPU, dim 4, the S^1 testbed, ≤ 5 seeds; the solvable core is quadratic and trained statements
316 are local at a critical point of V_θ , with anharmonic deviations measured as a zero-kick retention
317 bias of $+5$ to $+29\%$, kick-independent for 2/3 emergent seeds and amplitude-dependent for the
318 softest- μ^2 seed. The head-to-head is on *autonomous retention* in map-application units, and the unit
319 does not enter the input-driven task-RMSE axis. BIBO boundedness is evidenced within the coercive-
320 potential / compact-sublevel-set scope and concerns the autonomous, designed setting measured
321 here — nothing here claims that symplectic structure buys off-distribution stability in a *trained*,
322 *input-driven* setting, which we do not test and where a published symplectic architecture already
323 holds the ground with a gradient bound — and friction never stabilizes a saddle (Appendix E). The
324 symmetry framing is tree-level only. The finite- T diffusion law holds on both the designed channel
325 (verification) and the emergent arm (evidence, 3 seeds, above $T^* \approx 3 \times 10^{-3}$) under FDT-consistent
326 noise in a Newtonian kinetic mode; the one face that does not generalize is the continuous latch
327 register, which is designed-only. (i) *Scale is out of scope by choice*: every number here is dim 4,
328 hidden 64, ≤ 5 seeds, single-unit, laptop-CPU, and we report no result at larger dimension, wider
329 layers, more units or accelerator scale. (ii) *No external benchmark is won on its own headline metric*
330 *anywhere in this paper*; the comparisons are diagnostic retention protocols on a synthetic S^1 family
331 and matched-parameter ablations, not leaderboard results — and §4.3’s honest gap is part of the
332 claim, not a caveat to it. (iii) *The substrate scope, once, in our own voice: these laws govern the*
333 *store; end-to-end performance additionally depends on the encoder, measured separately, with its*
334 *parameter and state-byte budget ledgered.*

335 A price list changes what a flat direction is for. Read this way a trained memory has latches — exact
336 flat directions, the moduli of V_θ — and registers, nearly-flat modes of a chosen half-life, fenced as
337 designed-geometry in Appendix E; kinetic isotropy is then the price of an equivariant write *current*,
338 not of the register. Named as directions, not results: the μ^{-2} law, crossover and floor re-measured at
339 latent dimension $\gtrsim 64$ and $N \geq 8$ interacting units, where mode mixing can break the single-block
340 reduction that makes the price list exact here; the head-to-head on a wider trained-model family;
341 the solution-degeneracy instrument of Huang, Singh, Martinelli & Rajan (2025) applied to §4.3’s
342 designed-versus-emergent gap, which would separate what is recipe from what is architecture; the
343 anchored recipe at longer training and larger width; the equivariant-control wrapper that would put
344 the unit on the supervised task axis; a learned-store re-test of the explicitly-broken cell respecting
345 that fence; the map from (μ, γ, T) to lifetime whose $T = 0$ face is §4; and the price list re-derived
346 beyond abelian $SO(2)$, for a torus T^2 or the non-abelian $SO(3) \rightarrow SO(2)$ coset — a direction, on
347 which this paper offers no evidence.

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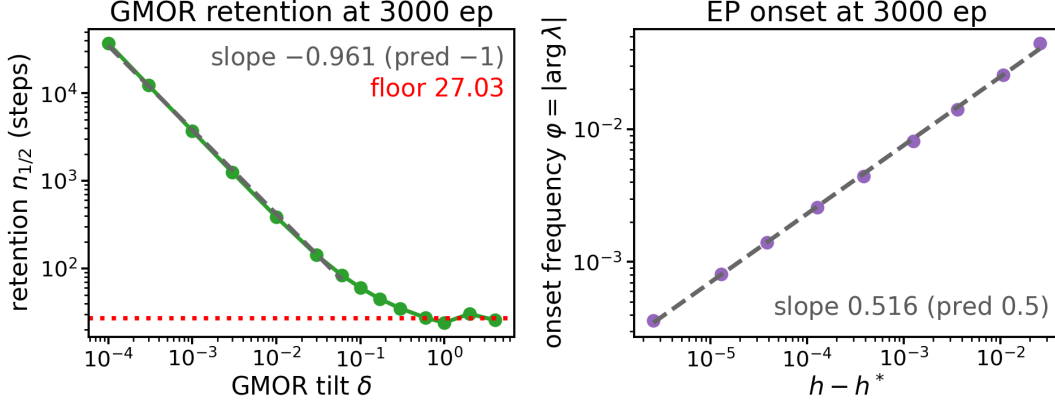


Figure 3: The headline laws of the price list survive the shipped anchor cure at 3000 epochs ($\approx 20\times$ the erosion horizon; anchored $\lambda=100$, 3 seeds). Left: the retention law $n_{1/2}$ versus tilt δ — anchored-3000-epoch points on the predicted slope, fitted -0.961 against predicted -1 , saturating at the curvature-independent floor 27.03 ; the μ^2/δ ratio is 1.0000 ± 10^{-12} over 4.6 decades. Right: the exceptional-point onset $\varphi \propto \sqrt{h - h^*}$ above the exceptional point, fitted slope 0.516 against predicted 0.5 , with $\varphi = 0$ exactly below — bit-identical to the 150-epoch value. *Designed testbed under the shipped cure; the survival of the laws through 3000 erosive epochs is evidence for the recipe.*

462 A The laws survive the training-time correction

463 B The price of the prior, the loan curve and the recovery ladder

464 *Demoted from the main text of this abstract, and retained in full: this appendix is a supporting result*
 465 *about what the physics prior costs, not one of this paper's claims.*

466 **The price of the prior (evidence).** Against param-matched controls — dim 4, 150 epochs, $\pm 0.05\%$
 467 params, 3 seeds, same MLP potential — the picture is Pareto, not podium. Volume conservation
 468 buys bounded (BIBO) dynamics within the coercive-potential / compact-sublevel-set scope (1.00
 469 versus 0.33 for a broken-volume arm), a protected near-flat direction (μ^2 0.008 versus 0.122) and
 470 vacuum survival under contrastive-divergence training ($r^* = 0.72$ versus 0); leapfrog buys the latch;
 471 and the prior costs raw fit, an unconstrained twin fitting $\approx 15\times$ better, 0.0128 against 0.190 eval
 472 MSE. That cost is *contraction-forbidden* by volume conservation — the broken-volume arm recovers
 473 $\approx 2.4\times$ of it precisely by leaking volume — and is not the causal velocity cap, inactive here. The
 474 loan is called at ≈ 700 steps (measured): the twin leads by 1–2 orders to ≈ 500 steps, crosses at
 475 ≈ 700 and diverges to 196 at 5000, while the unit holds a bounded plateau ≈ 0.20 – 0.23 . Its asset
 476 is boundedness by construction, not the lowest steady MSE — broken-volume (0.14) and LSTM
 477 (0.13) plateau below it — and of the licensed repairs only global damping buys the fit back, a learned
 478 friction *field* does not: paid mechanisms are not interchangeable. *The non-comparability caveat*
 479 *printed below travels with every number in this appendix.*

480 *Dim 4, hidden 64, learned-Newtonian kinetic term, dt = 0.05, circle $R=1\times 256$, 150 epochs, 3*
 481 *seeds, wake-only objective except the γ_ϕ rung. Cumulative fit MSE against horizon on the stationary*
 482 *circle-vacuum orbit, param-matched to $\pm 0.2\%$ (CLU 4549 / broken-volume 4557 / twin 4551).*

horizon (steps)	twin	CLU	broken-volume	LSTM
10	2.9×10^{-6}	2.2×10^{-5}	1.1×10^{-4}	7.5×10^{-2}
50	6.2×10^{-5}	1.2×10^{-3}	1.0×10^{-3}	4.9×10^{-2}
100	2.4×10^{-4}	1.2×10^{-2}	4.1×10^{-3}	4.9×10^{-2}
500	1.1×10^{-2}	1.8×10^{-1}	9.7×10^{-2}	7.7×10^{-2}
1000	3.2×10^{-1}	2.0×10^{-1}	1.2×10^{-1}	9.9×10^{-2}
5000	1.96×10^2	2.2×10^{-1}	1.4×10^{-1}	1.3×10^{-1}

The twin is best by 1–2 orders to ≈ 500 steps, crosses above CLU at ≈ 700 and diverges to 196 at 5000; CLU is the only arm flat *by construction*, on a bounded plateau ≈ 0.20 – 0.23 . Broken-volume (0.14) and LSTM (0.13) plateau slightly *below* CLU (0.22): “physics costs raw fit” persists as a small steady-state penalty, and CLU’s asset is boundedness, not the lowest plateau.

Non-comparability caveat, to be read before using these numbers. This ladder is a **separate wake-only experiment**, and its **absolute MSE scale and its twin (0.0047) are not cross-comparable** with the param-matched table above (twin 0.0128, different seeds, different training objective). **Only within-table gaps are meaningful.** In particular, do not divide across tables: the “ $\approx 15\times$ ” above (0.190/0.0128) and this table’s implicit “ $\approx 44\times$ ” (0.2066/0.0047) are the *same qualitative statement* measured on two different scales, not two different findings, and the 92% below is a fraction of **this** table’s absolute gap (0.202) only. The $\gamma = 0$ conservative rung (0.2066) is this experiment’s counterpart of the $\gamma = 0$ CLU of the param-matched table (0.190).

	rung	eval MSE	% gap recovered	BIBO	latch drift	flat μ^2
	twin (no physics)	0.0047	100% (def.)	1.00	—	—
497	CLU conservative ($\gamma = 0$)	0.2066	0% (def.)	1.00	0.778	5.2×10^{-3}
	$+\gamma$ global ($\gamma = 0.05$)	0.0216	92%	1.00	0.778	5.2×10^{-3}
	$+\gamma_\phi$ learned field ($K = 2$)	0.2548	−24%	1.00	0.706	2.1×10^{-2}

Each licensed mechanism is added one at a time; the gap recovered is $(\text{CLU}_{\gamma=0} - \text{rung}) / (\text{CLU}_{\gamma=0} - \text{twin})$, twin gap 0.202. Global γ recovers 92% of the **absolute** gap at zero structural cost, leaving BIBO, latch and μ^2 unchanged, while still trailing this table’s twin by $\approx 4.6\times$ in *ratio* (0.0216 against 0.0047); both statements must travel with the claim. The learned γ_ϕ field is a targeted-forgetting tool and does not recover fit, at -24% ; that rung retrained with the sleep phase on, so the “wrong tool” verdict is robust but the -24% is not a clean single-knob delta.

C The autonomous-retention head-to-head against learned baselines

The figure behind §4.3’s baseline paragraph; every number in it is stated there and none is added here.

D Per-step compute: CLU-Verlet against LSTM and LEM

Single-core CPU, f32, XLA; this backs the retirement of the “ $\approx 4\times$ longer” as a compute claim in §4.3. One CLU dissipative-Verlet step (kick–drift–kick, two $\nabla_q V$ backprops through a dim-4 $\rightarrow$ 64 $\rightarrow$ 64 $\rightarrow$ 1 tanh MLP, learned-Newtonian kinetic term, $\gamma=0.05$, $\text{dt}=0.05$) is measured against one LSTM and one LEM cell, both input 1, hidden 16, output 2, the exact retention baselines of §4.3. Wall time is the median of 7 scan-amortized repetitions over 2×10^5 steps; the naïve single-call timing is dispatch-bound at a $\approx 5 \mu\text{s}$ floor and is *not* reported.

	unit	params	FLOPs/step	wall ns/step (scan-amortized)
514	CLU Verlet (h64, dim 4, KDK)	4549	36148	1554
	LSTM cell (h16)	1186	2512	252
	LEM cell (h16)	1186	2400	500

	normalization	CLU vs. LSTM	CLU vs. LEM
515	raw map-steps (263 vs. 69/56)	$3.8\times$ longer	$4.7\times$ longer
	FLOP-normalized	$54.8\times$ more	$70.7\times$ more
	wall-normalized	$23.5\times$ more	$14.6\times$ more

Per-step ratios: CLU/LSTM = $14.4\times$ FLOPs and $6.2\times$ wall; CLU/LEM = $15.1\times$ FLOPs and $3.1\times$ wall. The $4\times$ **inverts** under compute normalization: per unit of retention the unit spends ≈ 15 – $24\times$ the wall time and ≈ 55 – $71\times$ the FLOPs of the RNN it out-holds. **Confound, flagged:** the

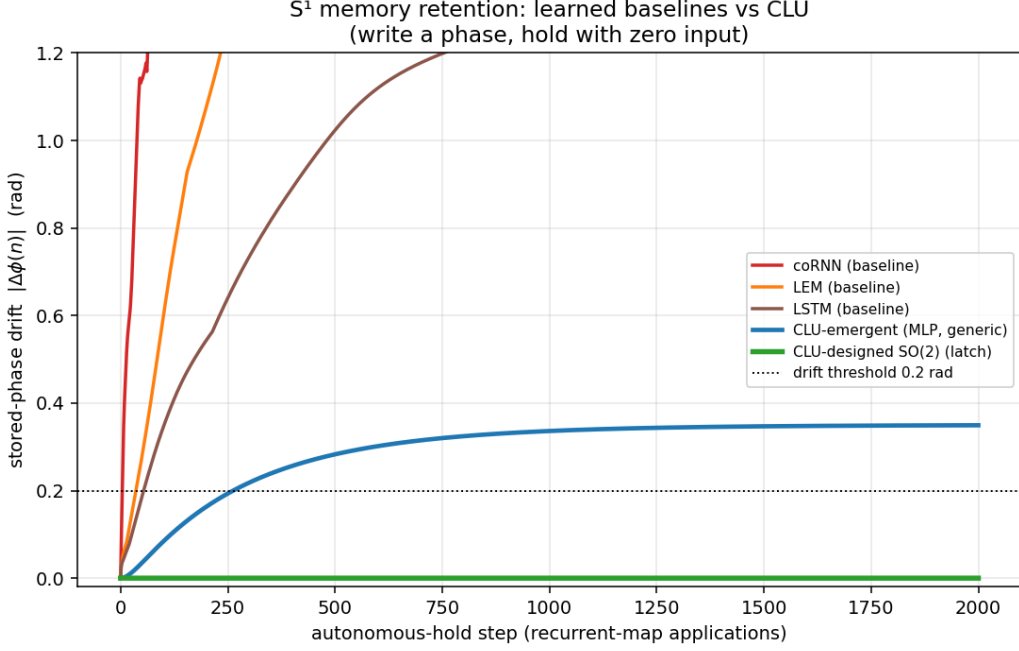


Figure 4: Autonomous retention on the S^1 protocol of the equivariant-Lyapunov preprint (arXiv:2605.03338): a phase is written, the input is removed, and the drift of the stored phase is tracked across recurrent-map applications. The learned baselines lose the stored phase within $\approx 5.6/56/69$ map-steps (coRNN/LEM/LSTM) and the stored coordinate randomizes past 1.2 rad; the generically-trained unit holds a bounded ≈ 0.35 rad and the symmetry-designed unit does not move. *Multi-seed status, stated plainly*: each baseline curve is the median over 5 seeds and the emergent curve the median over 3 seeds, while the designed curve is a single representative checkpoint — the 5/5-seed latch statement is §4.3’s, not this figure’s. Dim 4, hidden 64, laptop CPU, drift threshold 0.2 rad, $T = 0$. *Evidence*.

519 comparison is not width-matched, CLU at hidden 64 with two gradient evaluations against baselines
520 at hidden 16; a width-matched CLU would narrow the gap, but the sign is robust because the Verlet
521 step requires two potential backprops, and the 263/69/56 retention numbers come from these specific
522 configurations. The measurement is single-core CPU at batch 1; GPU or batched throughput could
523 shift the wall ratios, though not the FLOPs. §4.3 accordingly leads with the qualitative triad, states
524 the per-step $\approx 6\times$ (LSTM) and $\approx 3\times$ (LEM) wall factor, and does not carry the map-step ratio as a
525 compute claim.

526 E Prominent negatives

527 *Negatives are recorded, never dropped. Each row states the claim tested, the result, and the number*
528 *that settles it.*

	claim tested	result	number
	Does a learned inertial mass isotropize on symmetric data?	No; and the breaking is invisible to the Hessian and to the latch, showing only as a bounded Noether-charge oscillation.	Log-mass split ratios 0.98 / 1.18 / 5.4 from init to 150 ep; charge amplitudes $5.4/1.6/0.082 \times 10^{-2}$; tied controls exactly 1.0000.
	Does wake-sleep contrastive divergence preserve a designed degenerate vacuum?	No — it inverts it. The cure is a $V(\text{data})$ energy anchor, and the headline laws survive it (§4.4).	$r^* \rightarrow 0$ in 8/8 runs at 1000 ep; inversion at epoch 116/442/959 by sleep frequency; anchor $\lambda=100$ holds 5/5 seeds at $r^*=0.911 \pm 0.016$.
	Does an emergent (MLP) unit have a continuous coset register?	No — the continuous register is designed-only; an emergent unit stores a few bits on discrete washboard minima (evidence, 3 seeds; §4.3).	$1 - \lambda_{\text{coset}} \approx 10^{-3}$ against the designed $\leq 1.1 \times 10^{-15}$, ~ 12 orders; written δ retained $\leq 2.1 \times 10^{-3}$; capacity ≈ 1 –1.6 bits.
529	Do raw $n_{1/2}$ exponents discriminate designed from emergent?	No — instrument, not physics: a matched designed control on the same grid gives the same slopes, both carrying the ℓ_θ/Δ boundary layer. No $n_{1/2}$ may be quoted without its Δ and ℓ_θ/Δ .	Designed control $d \log n_{1/2}/d \log T = -0.53, -0.60, -1.04$ $d \log n_{1/2}/d \log \gamma = +0.78, +0.63, +0.55$; ℓ_θ/Δ up to 2.0.
	Is an explicit-breaking coefficient a lifetime dial on a learned store?	No, and the first half fails in sign: raising the breaking <i>lowers</i> the softest eigenvalue, and a coercivity term, not the breaking, sets the ceiling. A designed degeneracy does not survive superposition.	$\lambda_{\min}$ runs $+0.0994 \rightarrow -8.2846$ and $+0.3291 \rightarrow -1.1980$ over four decades of breaking; hard ceiling $\tau_{\max} = \Gamma/2\alpha = 4.0$; tilt-vacuum residual 0.140–0.343 against a random-direction baseline $1/\dim = 0.167$; tangential curvature predicted 0.100, measured 0.0994.
	Can this unit enter the input-driven path-integration task-RMSE axis?	No — it has no native velocity ingestion and the equivariant-control wrapper is unbuilt. No task-RMSE was fabricated.	—

claim tested	result	number
Do a learned friction field and a curvature governor compose usefully?	No — the pair is Pareto-dominated by either alone, and a learned compact-support gate under-covers without accurate placement.	Locus discovery 2/6 before adaptive- K , 2/3 after; best learned rejection 0.861; a λ re-sweep does not approach the oracle.
Can friction stabilize a saddle, or an energy gate catch isoenergetic escape?	No to both — a genuine scope limitation of the class. Only curvature control of V_θ and the mass-anisotropic cap $v_i^{\max} = c/\sqrt{M_i}$ bound expanding or isoenergetic modes.	$\lambda_+ > 1$ for all γ , with $\min_\gamma(\lambda_+ - 1) = 1.4 \times 10^{-5}$.
Is the coded momentum-coupled Langevin sampler FDT-exact in every kinetic mode?	No — exact only in the Newtonian modes; in the relativistic mode <i>no</i> noise scale gives the coded chain a Gibbs invariant at all.	Required per-mode scale $\sigma_i^* = \sqrt{M_i T \gamma (2 - \gamma)}$.
Is a mean-spectrum chaos regularizer informative?	No — it is degenerate; a chaos penalty must use a max or positive-part statistic.	—
Is sleep_temperature active at sleep_friction=0?	No — a silent knob; cite it when reproducing temperature-swept sleep runs.	$\sigma \propto \sqrt{\gamma_{\text{sleep}}}$, so the swept runs are bit-identical.

A reporting caution travels with the isotropization row. The closed-form charge-oscillation amplitude is an **on-vacuum-orbit** statement; **off the orbit only boundedness holds**. The amplitude-to-split ratio is moreover **not** constant, as the exact envelope requires, since it is linear only to leading order — so no proportionality constant should be quoted as a law, and we report amplitudes and boundedness, never a “drift rate”. The design rule attached to mass-tying is correspondingly about the *current*, not the register: **an anisotropic channel still latches with infinite half-life**; tie the masses for an angle-independent write current, and at $\dim(G/\mathcal{H}) \geq 2$ a degenerate pseudo-Goldstone multiplet, **not** to save the register.

Two reading rules travel with the learned-store row. (a) The breaking coefficient of that battery is an **explicit-symmetry-breaking magnitude**; it is **not** this paper’s integrator step ϵ , and the two must never be conflated. (b) No sentence in this paper may be read as claiming that a tilt magnitude is a lifetime dial on a learned store, or that a pseudo-Goldstone lifetime can be extended without bound by shrinking the breaking: on a store with a coercive confinement $\alpha\|q\|^2$ the ceiling is $\Gamma/2\alpha$, and lowering α breaks the write. This paper’s own tilt results (Appendix G) are analytic tilts on **designed** architecturally-invariant potentials, where the construction is exact; the learned-store scope is exactly what this negative removes. That battery is a *different architecture* from the trained units of this paper and shares none of their training configuration, which is why it is quoted as a scope fence and never as a measurement of the units here. What survives it is the *geometry*: the identity $\lambda_{\min} = \epsilon$ holds in the single-atom geometry in which it was derived, unit-tested to 5%, and the registered damped-mode floor $\tau = 2/\Gamma$ is confirmed, 5.0 predicted against 4.23–6.58 measured time units.

The sampler row’s scope, stated precisely, because the qualifier is load-bearing. The coded damping substep is an additive Gaussian momentum kick of fixed covariance, so the chain’s invariant momentum marginal is necessarily Gaussian-smoothed; for a *relativistic* kinetic term the Gibbs momentum marginal is Maxwell–Jüttner, which is not of that form (*theory note*, Prop-9’ — a characteristic-function argument needing neither $V = 0$ nor linear damping). **The failure is in the sampler, not in the thermodynamics**: the configurational equilibrium $\pi_q \propto e^{-V_\theta/T}$ is relativity-insensitive, since the momentum integral factorizes out, so a relativistic unit’s equilibrium distribution is perfectly well-defined and the chain simply fails to sample it; a one-line thermostat carrying a randomized, momentum-dependent inertia equal to the relativistic mass repairs it. We therefore never assert that a relativistic unit “has no equilibrium”. **This touches no result in this paper**: the trained

units here are `newtonian_learned` throughout, so σ^* is exact for every configuration we run — the qualifier is a *scope clause* on a class-level theorem, not a correction to a measurement.

F Positioning: the four retirements in full, and the head-to-head on the preprint’s own estimator

The four retirements of §2, elaborated so that every number behind them is on the record.

(1) Attention and state-space models have retention-guarantee analogues. HiPPO-LegS (Gu et al. 2020) has an $O(tL/\sqrt{N})$ approximation bound, a polynomial $\Theta(1/t)$ rather than exponential gradient decay, and *exact* timescale equivariance with no discretization step to tune — exact where ours is a discrete map at fixed ε with $h = \varepsilon\mu < 2$ a hard stability limit — and we have no measurement saying a learned curvature spectrum is preferable to it. Tunable permanence is likewise occupied (EDEN, Karuvally et al. 2025); UniCORN (Rusch & Mishra 2021b) is Hamiltonian, structure-preserving and *has* the gradient-bound theorem we lack; and the modern Hopfield update *coincides with* attention for a single head with identity projections and $\beta = 1/\sqrt{D}$ (Ramsauer et al. 2021).

(2) “Retrieval is a rollout, not a weighted sum” is not our novelty. The move is Kong, Brewer & Lai’s (2024), at capacity $N_c \propto K^{1.08 \pm 0.01}$ for their one-hot and binary index codings and 1.17 ± 0.02 for their 2D coding. Our differences are structural: neither symplectic, Hamiltonian nor mass-structured there, and the address enters through an index channel rather than as the initial condition of the state.

(3) Continuity is not an escape hatch from the NTM/DNC lineage. Both were already fully soft (Graves et al. 2014, 2016), so the precedent transfers as a warning. Of the three DNC failure modes Csordás & Schmidhuber (2019) diagnose, key/value entanglement, which they rank first, is architectural in a memory whose address q_0 is both the key and the initial condition of the value; de-allocation aliasing maps onto any position-gated forgetting; chained-read degradation is the third. None are measured here.

(4) Not a competitor to the transformer. Never claimed, and indefensible at this scale: $\approx 1\text{--}1.6$ bits per emergent register, no win over a trivial baseline on any external benchmark, and formal limits on fixed-size latent state for copying and retrieval (Jelassi et al. 2024) that conservation does not evade — it prevents *decay*, not *capacity*.

The head-to-head reproduced on the preprint’s own instrument. §4.2’s result does not depend on substituting the exact Jacobian gap for the preprint’s predictor. With the preprint’s own finite-horizon estimator $\hat{\lambda}(T=128)$ the result holds on that instrument: $\text{corr} = 0.9995$ overdamped, $\text{meas}/\text{pred} = 0.86\text{--}1.03$, and 0.30 at $\delta = 4$ against the exact-gap 0.31 .

G The price list verified in full, and GMOR proper on trained checkpoints

Designed testbed with analytic tilts and an analytic spurion, therefore verification of the theory’s exactness; 5 seeds at 150 epochs plus 3 anchored at 3000 epochs, dim 4, laptop CPU, f64 probes, probe-only with no retraining. The retention law $n_{1/2} \propto \mu^{-2}$ is a Gell-Mann–Oakes–Renner (GMOR) law in the spectral mass; we use the name for the identity $\mu^2 F^2 = \delta\Sigma$ throughout this appendix. GMOR proper — §G.3 onward — is demoted from the main text of this abstract and retained here in full; it is a supporting result, not one of this paper’s claims.

F.1 The price list, verified on trained models — the battery behind §4.1. §4.1 states this battery’s headline quantities: the constitutive identity to 3.2×10^{-10} over all 70 tilt rows on a machine-flat trained vacuum ($\mu^2 \leq 2.4 \times 10^{-15}$); the GMOR law $\mu_{\text{meas}}^2/\mu_{\text{pred}}^2 = 1.000000 \pm 5 \times 10^{-12}$ at every δ and seed across 4.5 decades; the fitted overdamped retention slope -0.985 ($n = 35$; predicted -1); the curvature-independent floor 27.03 steps at $\gamma = 0.05$; the metric bifurcation (0.01% agreement overdamped, a $3.2 \times$ split at $\delta = 4$); the frequency onset ($\varphi = 0$ exactly below h^* on 15/15 rows, fitted slope 0.5165 against $1/2$ above); and the latch transport law to $\leq 1\%$ over two decades of γ with the written angle frozen to $\leq 1.2 \times 10^{-15}$ rad. Four measurement details belong here rather than there. The $\sqrt{h - h^*}$ onset holds down to $h - h^* = 2.6 \times 10^{-6}$ on a fine multiplicative grid ($h - h^* \in [-4.2 \times 10^{-3}, 2.6 \times 10^{-2}]$); the slight slope excess over $1/2$ is far-field bending. The onset prefactor is 0.268 against the theory note’s 2×2 -normalized 0.3247, a one-line units reconciliation.

Where the quality factor permits ($f = 2, 4$) the trajectory frequency matches the Jacobian frequency to 0.06–0.3%; near onset the mode decays before one period completes, so the exceptional point is spectroscopically real but dynamically silent there. And the eigenvalue pair below h^* is real, which is what makes $\varphi = 0$ an exact statement rather than a resolution limit.

F.2 The damping corollary — a supporting result, deliberately not a headline. The same block fixes the half-life’s dependence on the damping, and that dependence is **not** monotone: sweeping γ at fixed spectral mass, $n_{1/2}(\gamma)$ traces a V-shape of depth $\approx 8\times$ whose argmin sits at, or just below, the exact critical-damping value $\gamma^* \approx 2\varepsilon\mu$, on all 5 designed seeds. Below γ^* the mode is underdamped at the floor $2\ln 2/(-\ln(1-\gamma))$, which *shortens* as γ grows; above it the mode is overdamped at $n_{1/2} \approx 2\gamma \ln 2/[(2-\gamma)(\varepsilon\mu)^2]$, which *lengthens*; the exceptional point is where the branches meet. Two consequences: **(i)** a retention measurement quoted at one γ is uninformative about the sign of $\partial n_{1/2}/\partial\gamma$ unless the band is named, the same architecture giving both signs; **(ii)** the $\mu \rightarrow 0$ latch corner is the limit $\gamma^* \rightarrow 0$, so a Goldstone register is permanently overdamped, which is the structural reason friction cannot erase it.

F.3 GMOR proper: why a second spurion. The angular tilt $V \rightarrow V + \delta(1 - \cos n\theta)$ of F.1 is *radius-independent* — measured on these checkpoints, $\max |r_{\text{tilt}}^* - r^*| = 2.22 \times 10^{-15}$ over all (checkpoint, δ) pairs — so it measures only the **product** $\mu^2 F^2 = \delta n^2$, in which the condensate degenerates to the pure number n^2 (measured: $\max |\mu^2 F^2/(\delta n^2) - 1| = 1.03 \times 10^{-11}$): the correct instrument for verifying a power law, and provably incapable of resolving the condensate. The **linear ambient spurion** $V \rightarrow V - \delta(u \cdot q)$ — the exact analogue of the ChPT quark-mass term, with u in the channel plane — lets the vacuum radius run with δ , so the three GMOR objects become separately measurable on a trained checkpoint: the pseudo-Goldstone spectral mass μ^2 from the autodiff Hessian at the polished vacuum; the decay constant $F^2 = M_{\text{ch}} r^{*2}$, the learned channel inertia times the vacuum radius squared and **not** the orbit radius; and the condensate $\Sigma = r^*(\delta) = -\partial E_{\text{vac}}/\partial\delta$. Over 8 trained checkpoints and $\delta \in [10^{-8}, 0.3]$ (10 values, 8 decades), $\max |\mu^2 F^2 - \delta\Sigma| = 1.33 \times 10^{-15}$ over all 80 pairs.

δ	$\Sigma = r^*$ (mean)	$\mu^2 F^2$ (mean)	max rel. dev.	max abs. dev.
10^{-8}	0.95186937111604	$9.5186937286 \times 10^{-9}$	1.10×10^{-8}	1.01×10^{-16}
10^{-7}	0.95186948174437	$9.5186948223 \times 10^{-8}$	1.87×10^{-9}	1.85×10^{-16}
10^{-6}	0.95187058802597	$9.5187058802 \times 10^{-7}$	1.87×10^{-10}	1.86×10^{-16}
10^{-5}	0.95188165067351	$9.5188165068 \times 10^{-6}$	2.18×10^{-11}	2.16×10^{-16}
10^{-4}	0.95199226031843	$9.5199226032 \times 10^{-5}$	1.34×10^{-12}	1.33×10^{-16}
10^{-3}	0.95309668422658	$9.5309668423 \times 10^{-4}$	2.27×10^{-13}	2.09×10^{-16}
10^{-2}	0.96398369127653	$9.6398369128 \times 10^{-3}$	2.69×10^{-14}	2.69×10^{-16}
3×10^{-2}	0.98732513581811	$2.9619754075 \times 10^{-2}$	9.66×10^{-15}	2.71×10^{-16}
10^{-1}	1.06435252346294	$1.0643525235 \times 10^{-1}$	1.55×10^{-15}	1.53×10^{-16}
3×10^{-1}	1.60757655208590	$4.8227296563 \times 10^{-1}$	1.55×10^{-15}	1.33×10^{-15}

Precision fine print, next to the claim. GMOR holds to machine precision in absolute terms at every δ ; relative exactness ($\leq 2.7 \times 10^{-14}$) is demonstrated for $\delta \geq 10^{-2}$. The growing relative deviation at small δ is **not** a failure of the law but the roundoff floor of the **autodiff Hessian**: the angular curvature is $K_{\theta\theta} = \delta/r^*$, reconstructed as a difference of $O(\|K\|)$ terms, so its absolute error is $\sim \epsilon\|K\|$ and its relative error $\sim \epsilon/\delta$. This was confirmed directly: on the analytic Mexican hat a **cancellation-free** Hessian returns relative deviation $1.1\text{--}2.2 \times 10^{-16}$ at every $\delta \in [10^{-8}, 0.3]$, while the autodiff Hessian on the same potential returns $2.28 \times 10^{-8} \rightarrow 4.22 \times 10^{-15}$ across that range, exactly $\propto 1/\delta$. **Do not quote “ 2.2×10^{-16} relative” for this experiment.** The same ϵ/δ floor applies to every μ^2 probe in this paper; all other sweeps here use $\delta \geq 10^{-4}$, where the floor is $\sim 10^{-12}$ and the quoted relative agreements are sound.

F.4 Σ is genuinely measured, three ways; it runs, and the angular tilt cannot see it. $\Sigma_{\text{geom}} = r^*(\delta)$; $\Sigma_{\text{HF}} = u \cdot q^*(\delta)$ by Hellmann–Feynman, agreeing to $\max |\Sigma_{\text{HF}} - \Sigma_{\text{geom}}| = 0.0$ exactly; and $\Sigma_{\text{FD}} = -dE_{\text{vac}}/d\delta$ by central finite difference at $h=10^{-6}$ — the envelope-theorem definition, structurally independent of the Hessian — agreeing to $\max |\Sigma_{\text{FD}}/\Sigma_{\text{geom}} - 1| = 9.2 \times 10^{-11}$. Under the linear spurion Σ runs by +16.1% to +210.1% across the δ grid (anchored-3000-epoch models +19.8%/+17.4%/+16.1%; designed-150-epoch models +210.1%/+169.3%/+35.9%/+34.4%/+41.1%), while under the angular tilt on the *same* checkpoints it does not move at all ($\leq 2.22 \times 10^{-15}$). With u

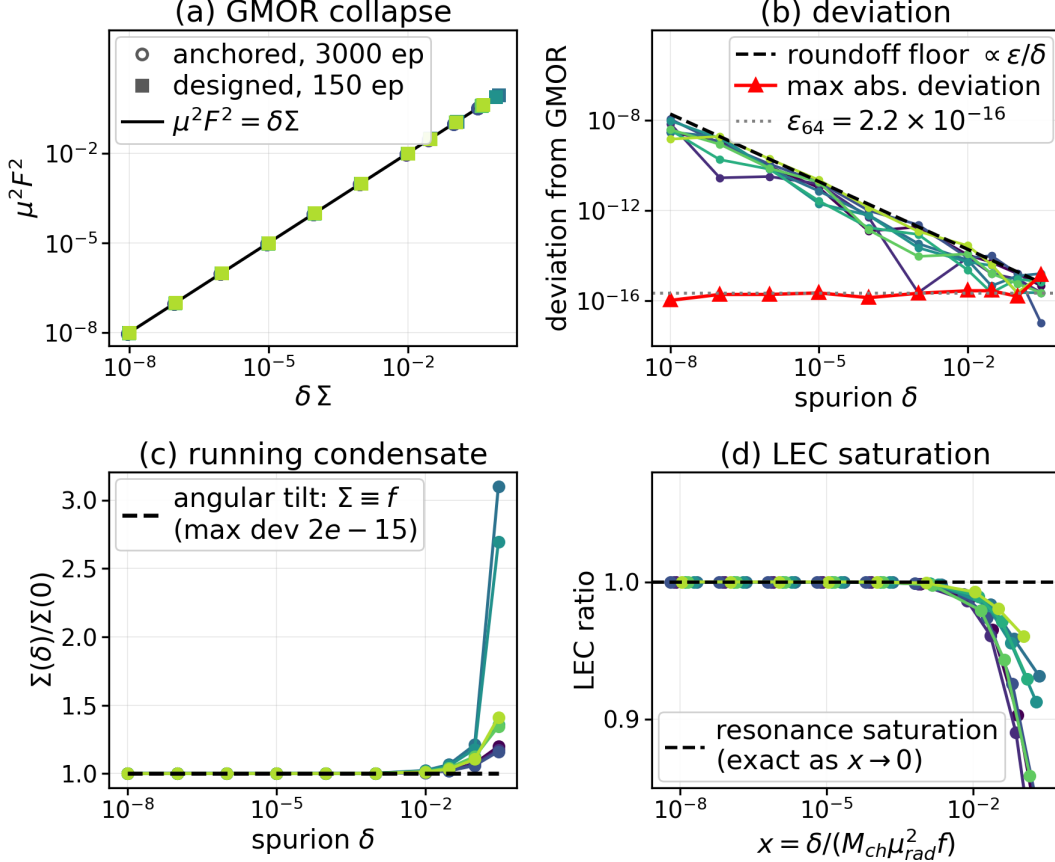


Figure 5: GMOR proper on trained checkpoints, with a linear ambient spurion. (a) The collapse $\mu^2 F^2 = \delta \Sigma$ over 8 decades of δ , 8 trained $SO(2)$ checkpoints. (b) Absolute deviation pinned at the f_{64} floor ($\leq 1.33 \times 10^{-15}$) against the ϵ/δ relative-roundoff line of the autodiff Hessian — the small- δ relative residual is a probe artifact, not a law violation. (c) The running condensate $\Sigma(\delta)/\Sigma(0)$ under the linear spurion against the angular tilt’s exactly flat $\Sigma \equiv r^*$ — the shipped tilt cannot see the condensate. (d) The low-energy-constant ratio tending to 1 as the chiral expansion parameter $x \rightarrow 0$, with slope -1.05 . *Designed testbed, analytic spurion, trained checkpoints, therefore verification.*

at channel angles $\{0.0, 0.7, -2.3\}$ on one designed checkpoint at $\delta=10^{-2}$, the μ^2 relative spread is 6.21×10^{-15} , as exact channel invariance requires.

F.5 The leading low-energy constant is saturated by the radial resonance. With $\mu_{LO}^2 \equiv \delta \Sigma(0)/F^2(0)$, the theory predicts a *relative* leading-order error $x \equiv \delta/(M_{ch} \mu_{rad}^2 f)$. Measured: the ratio is 0.99999607, the mean over 8 checkpoints at $\delta \leq 10^{-5}$, maximum deviation 2.20×10^{-5} ; from the measured μ^2 over $\delta \in [10^{-4}, 10^{-2}]$ it lies in $[0.97917, 0.99992]$; and corrections are first order, $d(\text{ratio})/dx = -1.0504$ at small x , i.e. ratio = $1 - O(x)$ — exactly what ChPT demands of a *leading* low-energy constant. The CLU therefore realizes ChPT’s **resonance saturation of low-energy constants** exactly rather than phenomenologically, the radial (σ) mode being the saturating resonance. **Breakdown is visible and expected:** at $\delta = 0.3$ the softest-radial seed, $\mu_{rad}^2 = 0.670$, has $x = 0.68$ — not small — and Σ has run +210%; the expansion is uncontrolled there, so $\delta = 0.3$ **must not be quoted for the next-to-leading-order claim:** cap at $x < 0.25$ or state x explicitly.

F.6 Honest scope. This appendix is tree-level and classical: no loops, chiral logarithms, running low-energy constants or anomalies; “resonance saturation” is an exact algebraic statement about a two-mode classical potential, not a phenomenological fit. It is probe-only on already-trained checkpoints, spurion off by default so as to be bit-compatible with every result elsewhere, on one architecture family — designed exact $SO(2)$, dim 4, laptop-CPU. The emergent MLP arm is untested and would carry its own self-breaking δ_{eff} , so GMOR there should be tested as $\delta \rightarrow \delta + \delta_{eff}$: an open falsifiable, not a result.